

Three-dimensional atom localisation with calibrated uncertainty from multislice electron ptychography: mapping polar order in a $\text{PbTiO}_3/\text{SrTiO}_3$ superlattice

Tristan McCarthy,¹ Peng Wang,² and Nicholas D. M. Hine²

¹*HetSys Centre for Doctoral Training, University of Warwick, Coventry CV4 7AL, UK*

²*Department of Physics, University of Warwick, Coventry CV4 7AL, UK*

(Dated: August 15, 2026)

Polar vortices in ferroelectric superlattices are intrinsically three-dimensional: the polarisation rotates along the electron beam as well as across it, so a projection image integrates away the very structure it is used to measure. Multislice electron ptychography (MEP) recovers depth from a single orientation, but converting its phase volume into atomic coordinates is obstructed by an extremely anisotropic response — 0.1 Å in-plane against 1–2 Å along the beam — which buries the oxygen sublattice under its titanium neighbours. We simulate 4D-STEM of a relaxed $\text{PbTiO}_3/\text{SrTiO}_3$ supercell, invert it by MEP, and localise every atom by fitting the volume as a superposition of a *measured* single-atom response. Against the known structure the method recovers 95.5% of bulk oxygen with 1.1% species confusion, where the established route — three-dimensional deconvolution followed by peak detection — reaches at most 59%; the entire advantage lies in the axially overlapped oxygen (recall 82% against 9–26%). Per-atom uncertainties combine a *joint* Cramér–Rao bound, which corrects the twofold variance inflation that a per-peak bound misses at the 1.95 Å Ti–O spacing, with distribution-free Mondrian conformal calibration; the intervals hold their nominal coverage to within 1%, and the procedure carries over with no retuning to an independent reconstruction at a different sampling and dose. Propagating them through the Ti–O₆ off-centring gives the local polarisation with error bars: the in-plane component is recovered to a median 0.007 Å and 0.9°, while the along-beam component is not measured at all — and the calibrated uncertainty reports this without recourse to ground truth. First-principles core-hole DFT shows that the missing component is nevertheless spectroscopically accessible: the O K edge is 78% dichroic with respect to the polar-axis orientation, of which a monotonic 19% is the ferroelectric displacement’s own signature, though recovering it requires a collection aperture $\lesssim 2$ mrad rather than the 100 mrad ptychography optics.

I. INTRODUCTION

Ferroelectric oxide superlattices host polar textures — vortices, skyrmions, labyrinths — that have no counterpart in bulk ferroelectrics and are now a mature field of their own [1–4]. In $\text{PbTiO}_3/\text{SrTiO}_3$ (PTO/STO), competition between elastic, electrostatic and gradient energies drives the local dipoles into rotational arrangements whose topology, not merely whose magnitude, is the physical object of interest. Establishing that topology experimentally means measuring a vector field at atomic resolution, and the standard tool for doing so is atomic-resolution STEM: the polarisation is inferred from the sub-ångström offset of the B-site cation from the centre of its oxygen cage, mapped column by column.

That measurement is a projection. A conventional STEM image integrates along the beam, so it returns the component of the off-centring lying in the image plane and is blind to the component along the viewing direction. For a uniformly polarised domain this is a minor restriction, cured by imaging a second zone axis. For a vortex it is not: the polarisation *rotates with depth*, so different depths within a single atomic column carry different — sometimes opposing — dipoles, and the projected image reports their average. The structure that defines the texture is exactly the structure the measurement removes.

Multislice electron ptychography (MEP) offers a way

out. By recording the full diffraction pattern at every probe position and inverting a multislice forward model, MEP reconstructs the specimen as a stack of slices rather than a single projection, achieving optical sectioning from a *single* orientation [5–7]. Its lateral resolution is set by the illumination angle rather than the lens [8–10], and it has matured rapidly: dopant depths in bulk crystals [6], oxygen vacancy sites in nickelates [11], sub-nanometre depth resolution from a few coupled small-angle projections [12]. The alternative, atomic electron tomography, gives three-dimensional coordinates with quantified precision [13–15] but needs a large tilt series — prohibitive for a beam-sensitive oxide, and incompatible with the single-orientation acquisition a vortex measurement must be built on.

The obstacle is what happens *after* the reconstruction. A single-orientation measurement is aperture-limited in depth: for convergence semi-angle α the lateral resolution scales as $\lambda/2\alpha$ but the axial as λ/α^2 , which for the 300 keV, 100 mrad probe used here are ≈ 0.10 Å and ≈ 2.0 Å respectively. The reconstructed response of a single atom is therefore more than an order of magnitude longer along the beam than across it. Since that length exceeds the 1.95 Å Ti–O spacing of the perovskite sublattice, neighbouring responses overlap: an oxygen atom sits inside the axial flank of its titanium neighbour. Local peak detection cannot separate them, and the established remedy — three-dimensional Richardson–Lucy or

maximum-entropy deconvolution of the volume followed by peak analysis [16] — was developed for sparse, isolated features rather than for a dense sublattice. Meanwhile the two-dimensional community long ago settled on model-based least-squares fitting of a peak superposition, whose defining virtue is that it handles overlap explicitly [17]. That machinery has not been carried into three dimensions for every atom of a dense lattice at once.

There is a second gap, and for a physical conclusion it is the more serious one. A polarisation map derived from located atoms is a difference of coordinates, each carrying an error; whether a measured off-centring is real or is an artefact of localisation noise cannot be judged without a per-atom uncertainty that is *calibrated*, not merely reported. Reconstruction pipelines rarely provide one. A least-squares covariance is optimistic on near-noiseless simulated data, where the error is systematic rather than statistical, and the natural per-peak Cramér–Rao bound is the bound with all neighbours known exactly — precisely the assumption that fails on an overlapped lattice.

This work addresses both gaps and applies the result to the physics. We (i) simulate 4D-STEM of a relaxed PTO/STO supercell and reconstruct it by MEP; (ii) localise every atom by fitting the phase volume as a superposition of a *measured* single-atom response, extending model-based quantification [17] to three dimensions and to all species at once; (iii) attach per-atom uncertainties that combine a joint Cramér–Rao bound with distribution-free conformal calibration, and demonstrate that they transfer to an independent reconstruction; (iv) propagate them into a Ti–O₆ polarisation map, showing quantitatively which component of the polarisation the measurement does and does not determine; and (v) ask, by core-hole DFT, whether the component ptychography misses is recoverable spectroscopically. Throughout, the localisation never sees the ground truth: known coordinates are used only to score the result.

II. METHODS

A. Model system

The test object is a relaxed atomistic supercell of a PTO/STO superlattice provided by collaborators — $70.0 \times 70.0 \times 47.9 \text{ \AA}$, 19440 atoms, 1944 formula units of each material. Both are perovskites (ABO₃): heavy Pb or Sr on the A site, Ti on the B site, oxygen on an octahedral cage. In the PTO layers the B-site titanium and A-site lead displace from their centrosymmetric positions, and the resulting dipoles order into the labyrinth pattern characteristic of these superlattices. We quantify the local polarisation throughout by the B-site off-centring

$$\delta_i = \mathbf{r}_{\text{Ti},i} - \frac{1}{6} \sum_{j \in \text{O}_6(i)} \mathbf{r}_{\text{O},j}, \quad (1)$$

the displacement of each titanium from the charge centre of its six nearest oxygens. Figure 1(a) maps its in-plane projection over the whole cell.

This is an exacting test on three counts: the scattering contrast spans two orders of magnitude, from lead ($Z = 82$) to oxygen ($Z = 8$), so the light sublattice sits near the noise floor; the signal is a sub-ångström displacement of atoms 1.9–3.9 Å apart; and the polarisation varies along the beam as well as across it, twisting with depth into a vortex tube within the scanned volume (Fig. 1c). The beam is set along a pseudocubic $\langle 100 \rangle$ direction, a $\approx 70 \text{ \AA}$ path; the known input coordinates are the ground truth against which every result is scored.

B. 4D-STEM simulation

The simulation is the forward operator $\mathcal{S}$ mapping the structure to the data a microscope would record, $\mathbf{I} = \mathcal{S}(\{\mathbf{r}_a, Z_a\}) + \text{noise}$. A convergent probe of semi-angle $\alpha = 100 \text{ mrad}$ at 300 keV ($\lambda = 1.969 \text{ pm}$), focused 20 Å above the entrance surface, is propagated through the specimen by the multislice recursion [18, 19]

$$\psi_{n+1} = \mathcal{P}_{\Delta z} \otimes [t_n(\mathbf{r}) \psi_n(\mathbf{r})], \quad t_n = \exp[i \sigma v_n(\mathbf{r})], \quad (2)$$

with v_n the projected potential of slice n , σ the interaction constant and $\mathcal{P}_{\Delta z}$ the Fresnel propagator; the recorded quantity is $I(\mathbf{r}_p, \mathbf{k}) = |\mathcal{F}\psi_N|^2$. We use abTEM [20] on GPU with Lobato potentials [21]. Thermal motion enters in the frozen-phonon approximation [22] as an incoherent average over 16 configurations, with displacement amplitudes set *per species* from tabulated Debye–Waller factors [23]: lead and oxygen vibrate roughly twice as far as titanium ($B = 0.90$ and 0.80 against $0.45 \text{ \AA}^2$), so a single averaged amplitude would misweight the diffuse background of exactly the species hardest to detect. A finite electron budget is imposed as shot noise, $n \sim \text{Poisson}(N_e \tilde{I})$, over $D = 10^4\text{--}10^{10} \text{ e \AA}^{-2}$.

C. Multislice ptychographic reconstruction

Reconstruction inverts $\mathcal{S}$ by iterative ptychography [24, 25]. To resolve structure along the beam the single multiplicative projection is replaced by a stack of N_L transmission slices [7]; slices and probe are recovered jointly by minimising the negative log-likelihood of the shot-noise model, implemented as the variance-stabilised amplitude metric [26] in the LSQ-ML engine of Odstrčil *et al.* [27] within PtychoShelves [25]. Depth information is available because the overfocused probe propagates dynamically through a specimen many times thicker than its depth of focus and the reconstruction recovers that propagation slice by slice — optical sectioning, not tomography. We reconstruct $N_L = 70$ slices (coherent, $\Delta z_{\text{rec}} \approx 1.0 \text{ \AA}$) or 105 (thermally averaged, $0.67 \text{ \AA}$). Critically, *no regularisation is applied to the inter-layer coupling*: the axial structure is set by the data rather than by

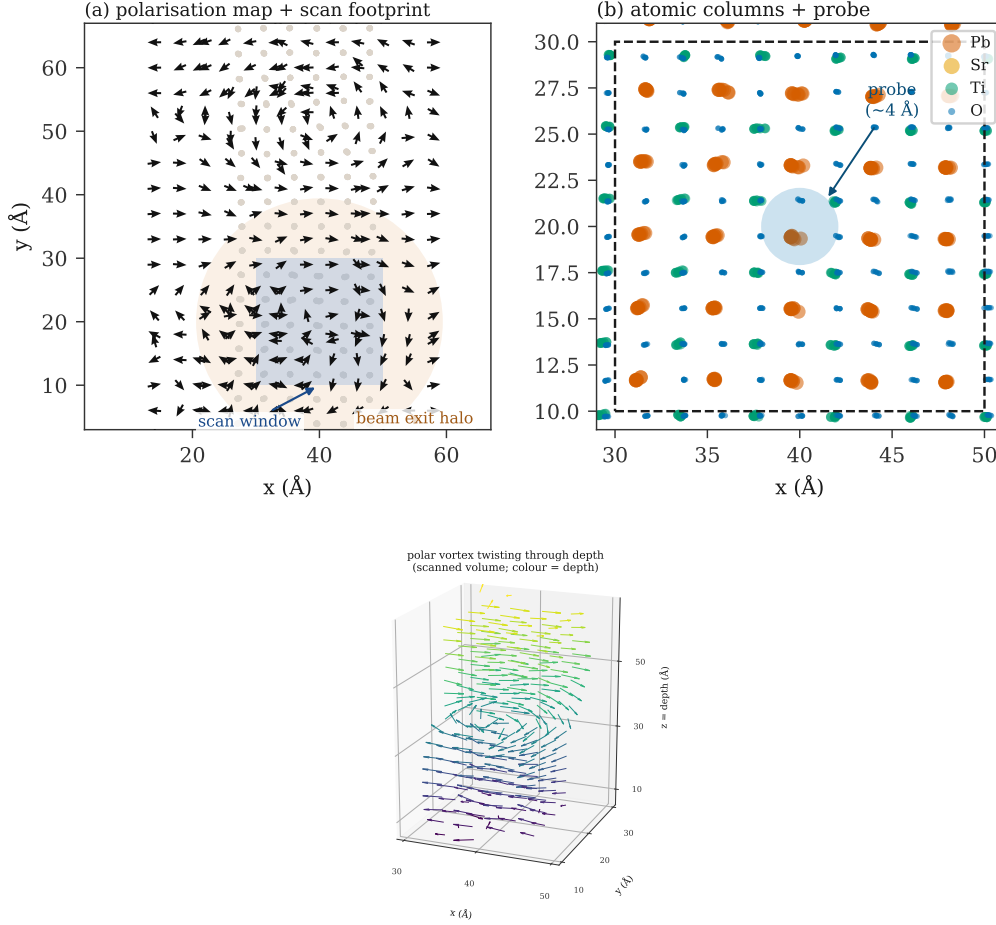


FIG. 1. The model system. (a) In-plane polarisation over the whole cell viewed down the beam, one mean-direction arrow per atomic column, resolving the polar domains and vortices. Shaded square: the 20 Å scan window; shaded disc: the exit-wave footprint the recorded diffraction spans, which is why the full box is simulated. (b) Atomic columns in beam projection within the scan region with the overfocused probe (≈ 4 Å) overlaid; the lead columns split into polar dumbbells. (c) The polarisation of the scanned volume in three dimensions: the local dipoles (arrows, coloured by depth) twist into a vortex tube along the beam — the depth variation that a projection integrates away.

a smoothness prior, which would bias the very coordinate we go on to quantify. For the dosed data the illumination is treated as a mixed state [28] with orthogonal probe relaxation [29], so that the thermal diffuse background is absorbed by incoherent degrees of freedom rather than forced into the coherent object. All parameters are collected in Table I.

D. Atom localisation

a. Forward model. We treat the reconstructed object as a sparse superposition of identical single-atom responses,

$$\phi(\mathbf{r}) = \sum_{i=1}^N \beta_i K(\mathbf{r} - \mathbf{r}_i) + b(\mathbf{r}) + \varepsilon(\mathbf{r}), \quad \beta_i \geq 0, \quad (3)$$

with ϕ the per-layer median-subtracted phase, $(\beta_i, \mathbf{r}_i)$ the amplitude and three-dimensional position of atom i , and b a slowly varying background removed by a high-pass filter. The reconstruction is a nonlinear inverse problem, so Eq. (3) is an empirical linearisation rather than a derived identity; we test it directly in Sec. III A.

The kernel K is *measured*, not parameterised: a single isolated atom is propagated through the byte-identical simulation and reconstruction pipeline and its reconstructed response is taken as K . This matters because K is strongly anisotropic and its axial profile is asymmetric, so no simple parametric form is faithful. We use the lead kernel for all species — an isolated oxygen or titanium does not reconstruct above the noise floor, so no measured light-atom kernel exists — on the argument that the response *shape* belongs to the imaging system while the amplitude β carries Z .

b. Support selection. With K normalised to $\|K\|_2 = 1$, the least-squares amplitude of a single source at $\mathbf{r}$ is

Stage	Parameter	Value
Sample	composition supercell	PbTiO ₃ /SrTiO ₃ , 19 440 atoms 70.0 × 70.0 × 47.9 Å
Probe	energy / α defocus	300 keV / 100 mrad −20 Å (20 Å overfocus)
Scan	window / step	20 Å / 0.10–0.15 Å
Sim.	slice thickness frozen phonons detector dose	2.0 Å (0.5 Å fine) 16 configs; per-species B 200 mrad, binned to 356 ² px 10 ⁴ –10 ¹⁰ e Å ^{−2} (Poisson)
Recon	engine layers N_L regularisation	PtychoShelves LSQ-ML 70 / 105 ($\Delta z_{\text{rec}} = 1.0$ / 0.67 Å) inter-layer off; $\beta_{\text{LSQ}} = 0.1$
DFT	code / functional core hole supercell spectra	CASTEP [30] / PBE full (1s ¹ /2p ⁵), charge +1 2 × 2 × 2 (40 atoms) OptaDOS [31], $\hat{\mathbf{q}}$ -resolved

TABLE I. Simulation, reconstruction and first-principles parameters.

the matched filter $(K \star R)(\mathbf{r})$ applied to the current residual R , and the position maximising it maximises the achievable reduction in residual sum of squares. Selection is therefore greedy forward selection on the dictionary $\{K(\cdot - \mathbf{r})\}$ — matching pursuit [32], equivalently the CLEAN algorithm of radio astronomy [33] — with a minimum-separation exclusion enforcing identifiability. Iteration stops when $\max(K \star R) < k \sigma_{\text{MF}}$, where σ_{MF} is a median-absolute-deviation estimate of the matched-filter noise over the sub-median voxels of the same sub-volume and $k = 2$ throughout. The threshold is noise-relative rather than absolute because $(K \star R)$ scales with the phase amplitude of the reconstruction, which differs by a factor of about two between the two reconstructions used here; an absolute threshold does not transfer between them.

c. Refinement. Writing $\theta_i = (\beta_i, \mathbf{r}_i)$, we minimise $\|\phi - \sum_i \beta_i K(\cdot - \mathbf{r}_i)\|^2$ by block-coordinate descent: each atom is updated in turn by Gauss–Newton steps on the local residual *with the current models of all other atoms subtracted*, so overlap between neighbours is accounted for to first order at every update. Derivatives of K are obtained by finite differences on the measured kernel. A candidate is retained only if the normalised correlation between the local data (neighbours removed) and its fitted response exceeds a fixed threshold — a goodness-of-fit criterion on *shape*, independent of amplitude, so that weak but well-formed responses are not discarded along with the poorly formed ones. This is model-based fitting of a peak superposition [17] carried into three dimensions with a measured rather than parametric kernel; estimating depth by fitting the axial response is established for isolated dopants [6, 34], and is applied here to every atom of a dense lattice.

d. Species and lattice-constrained detection. Columns are typed from their own fitted-amplitude statistics, which are well separated in this material; within mixed B–O columns the two sublattices are separated by amplitude, with ambiguous cases resolved by consistency with the local periodicity of already-assigned neighbours. Sites predicted by that periodicity — from the fitted atoms of the same column, never from the reference structure — but left empty by selection are re-examined at a reduced detection threshold with bounded positional and amplitude tolerances. This deliberately trades false negatives against false positives for the weakest species only; such detections are flagged in all outputs so that the constrained and unconstrained populations can be reported separately, and every number below is given for both.

E. Uncertainty quantification

We separate a *model* standard error, computed without ground truth, from a *calibration* step that converts it into intervals with a stated coverage.

The model error has two terms, $\sigma^2 = \sigma_{\text{stat}}^2 + \sigma_{\text{sys}}^2$. The statistical term is the Cramér–Rao bound of the *joint* estimation problem: for the stacked parameter vector $\theta = (\theta_1, \dots, \theta_N)$ the Fisher information of Eq. (3) under homoscedastic residuals is $\mathbf{F} = \mathbf{J}^\top \mathbf{J} / s^2$, and the attainable variance of θ_i is the corresponding diagonal block of $\mathbf{F}^{-1}$, *not* the inverse of the diagonal block $\mathbf{F}_{ii}$. The two coincide only for an orthogonal design; in general $[\mathbf{F}^{-1}]_{ii} \geq [\mathbf{F}_{ii}]^{-1}$ and the ratio is the variance inflation factor (VIF) of the overlapping templates. Evaluating $[\mathbf{F}_{ii}]^{-1}$ — each atom fitted with its neighbours held fixed — is the per-peak bound used to predict single-column and single-dopant precision [6, 17]; on a dense sublattice it is the bound with all neighbours known exactly, and it is optimistic. We therefore assemble $\mathbf{J}$ over all atoms in a subvolume and invert $\mathbf{F}$ once (a $4N \times 4N$ system with $N \leq 45$), taking each atom’s marginal variance. The systematic term σ_{sys} is the position spread induced by kernel mismatch, estimated by fitting a synthetic response generated with one measured kernel using another admissible kernel; it requires no ground truth and so transfers to experimental data.

Intervals then follow by split-conformal calibration [35, 36]: on a volume with known positions we form non-conformity scores $s = |\Delta|/\sigma$ per axis and take the empirical $(1 - \alpha)$ quantile with the finite-sample correction, reporting $q_{1-\alpha}\sigma$ as the half-width. This attains the target coverage by construction with no Gaussian assumption. Because the error is strongly heteroscedastic across the lattice we calibrate in the Mondrian (label-conditional) variant [37], stratifying by species, detection mode (unconstrained versus lattice-constrained) and depth band, so that coverage holds *per stratum* rather than only on average. The 95% interval is the default in released coordinates; the calibration table and per-stratum coverage

are exported with the data, so a dataset on which the calibration fails to transfer is identified rather than silently trusted.

F. Polarisation from located atoms

Equation (1) is evaluated on the located atoms exactly as on the model: for each located titanium, the six nearest located oxygens within $2.8 \text{ \AA}$ form the cage, and δ is the offset of the titanium from their centroid. Titanium whose cage is incomplete in the reconstruction is excluded and reported separately rather than completed from the lattice. Uncertainty on δ is propagated by Monte Carlo (400 draws) from the per-atom conformal half-widths of Sec. II E — that is, from quantities available *without* ground truth, which is what a downstream user actually holds.

G. First-principles EELS

In core-loss EELS the beam electron ionises a core state; the energy-loss near-edge structure (ELNES) is governed by the dynamic form factor entering the double-differential cross-section [38]. For the small momentum transfer $\mathbf{q}$ of a near-axis collection the dipole expansion gives

$$S(\mathbf{q}, E) \propto q^2 \sum_f |\langle f | \hat{\mathbf{q}} \cdot \mathbf{r} | i \rangle|^2 \delta(E - E_f + E_i), \quad (4)$$

so the ELNES measures the symmetry-projected unoccupied local density of states resolved along $\hat{\mathbf{q}}$, which for a small on-axis aperture is essentially the beam direction. In an anisotropic crystal this depends on the orientation of $\hat{\mathbf{q}}$ relative to the lattice [39] — here, relative to the polar axis. Two consequences frame the study. First, because $|\langle f | \hat{\mathbf{q}} \cdot \mathbf{r} | i \rangle|^2$ is even in $\mathbf{q}$, the dipole ELNES sees the *magnitude* of the along-beam polarisation but not its *sign*. Second, a real measurement integrates over the convergence and collection apertures, and the anisotropic term is weighted by a geometric factor vanishing at the *magic angle* $\beta^* \approx 4\theta_E$, with $\theta_E = E/(\gamma m_0 v^2)$ [40].

We evaluate Eq. (4) with the plane-wave pseudopotential code CASTEP [30] using the core-hole method [41]: the ionised atom carries an on-the-fly-generated pseudopotential with one fewer core electron (a full core hole) plus a neutralising background, in a $2 \times 2 \times 2$ supercell that dilutes the hole from its periodic images, so the conduction states are computed in its presence. The core-to-conduction matrix elements are contracted by OptaDOS [31, 42] into the core-loss spectrum with $\hat{\mathbf{q}}$ set explicitly. The central quantity is the ELNES dichroism

$$\Delta(E) = S(\mathbf{q} \parallel c, E) - S(\mathbf{q} \perp c, E), \quad (5)$$

the difference between probing along and across the polar axis, which by reciprocity is the change in the measured

edge when the polarisation rotates from along the beam to in-plane. We take the O K edge as the primary observable — well described at the single-particle level and an established probe of ferroelectric off-centring [43, 44] — and treat Ti $L_{2,3}$ for anisotropy trends only, since its $2p$ spin-orbit multiplet structure is not captured by single-particle DFT [45]. Calculations run on the Warwick Blythe cluster; each PbTiO_3 core-hole SCF takes 40–45 min on 32 cores and each OptaDOS core-loss pass 75–90 min.

III. RESULTS AND DISCUSSION

A. The measured response and the forward model

The reconstructed single-atom response (Fig. 2a) has full width at half maximum $0.10 \text{ \AA}$ in-plane against $1.0 \text{ \AA}$ along the beam for the coherent reconstruction and $2.0 \text{ \AA}$ for the thermally averaged one — the anisotropy the single-orientation aperture scaling predicts, with the coherent figure a little sharper than the $\delta z \approx \lambda/\alpha^2$ estimate on noiseless data. Because that axial width is comparable to or larger than the $1.95 \text{ \AA}$ Ti–O spacing, neighbouring responses overlap and the design is not separable.

The decisive question is whether the apparent axial blur is *overlap of a fixed response* or a genuine loss of information. We test Eq. (3) directly by fixing $\{\mathbf{r}_i\}$ at the known positions and solving for $\{\beta_i\}$ only. A comb of measured kernels at the true depths reproduces the observed depth profile of a mixed B–O column at a median $r = 0.94$ over 20 columns (Fig. 2b), and $r = 0.92$ in the full depth- x cross-section. The blur is therefore kernel overlap, not lost information — which is why a method that models the overlap can beat one that does not.

B. Localisation performance

Table II scores the method against the known structure, alongside the established alternatives run on the *same* volume with the *same* measured kernel, each at its own best-effort threshold. Over the bulk ($z = 10\text{--}56 \text{ \AA}$, excluding the entrance and exit artefact bands) the model fit recovers 95.5% of oxygen, 96.4% of titanium and 95.6% of lead, with 1.1% of species labels wrong, in-plane RMS accuracy $0.032 \text{ \AA}$ and depth RMS $0.37 \text{ \AA}$.

The aggregate oxygen numbers understate what is actually being tested, so we split the oxygen sublattice by its geometry. Oxygen that is *in-plane isolated* — sitting in its own column, resolved laterally — is found by every method, including raw peak-picking, at 79–92%. Oxygen that is *axially overlapped* — sharing a column with titanium $1.95 \text{ \AA}$ away, hidden inside the flank of its neighbour’s response — is the hard case, and it is where the methods separate completely: 82% for the model fit against 26% for raw peak-picking, 13% for maximum-entropy deconvolution and 9% for

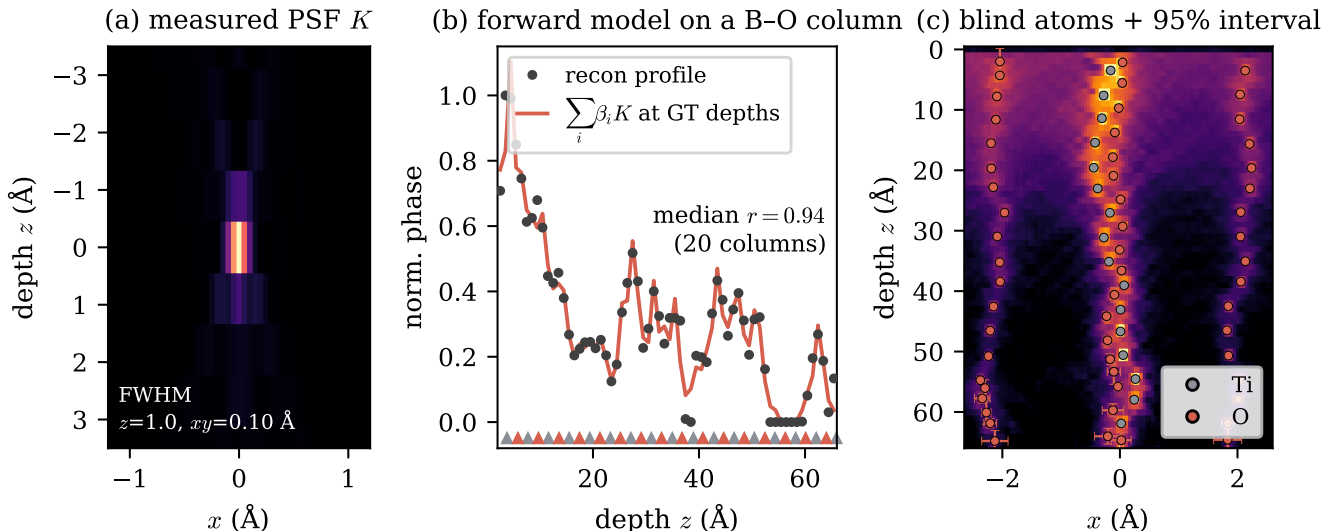


FIG. 2. Model-based localisation. (a) The measured point-spread function K (depth- x slice): tight in-plane, elongated along the beam by the aperture-limited axial resolution of a single-orientation measurement. (b) Direct test of the forward model, Eq. (3): the measured depth profile of a mixed Ti-O column against the superposition $\sum_i \beta_i K$ of the measured response at the *known* atomic depths, matching at a median $r = 0.94$ over 20 columns. Triangles mark the true Ti (grey) and O (red) depths. The apparent 2–4 Å axial blur is overlap of a fixed response, not a loss of information. (c) Blind atoms located on a raw cross-section, coloured by assigned species, with calibrated 95% intervals.

method	recall (%)			oxygen split (%)	
	Pb	Ti	O	isolated	overlapped
3-D peak-pick, raw	93	88	72	91	26
RL deconv. + peak-pick	93	87	58	79	9
MEM deconv. + peak-pick	93	89	59	79	13
1-D per-column deconv.	91	65	49	62	20
3-D model fit	88	89	89	92	82
unconstrained only	88	89	86	92	71
bulk, $z = 10\text{--}56$ Å	96	96	96	—	—

TABLE II. Recovery of each sublattice, whole field unless stated, on the same reconstruction with the same measured kernel. “Isolated” oxygen occupies its own column; “overlapped” oxygen shares a column with titanium 1.95 Å away. Precision is 0.97 for the model fit (0.98 unconstrained) against 0.84–0.99 for the baselines; only the model fit returns species labels and per-atom uncertainties.

Richardson–Lucy. This is the entire advantage of the approach, and it is exactly the population that a projection-blind, overlap-limited measurement must recover if the oxygen cage is to be reconstructed at all.

Two negative results deserve emphasis because they run against expectation. First, **deconvolution before detection does not help, and Richardson–Lucy actively hurts the oxygen**: both restoration routines *lose* oxygen relative to peak-picking the raw volume (58–59% against 72%). They redistribute dynamic range towards the strongly scattering species, so weak oxygen falls below any relative floor, and they cannot restore the ax-

ial spatial frequencies a single-orientation measurement never records — the frequencies that separate an oxygen atom from the flank of its titanium neighbour. Ishizuka *et al.* report maximum entropy as the stronger routine [16]; we reproduce no meaningful advantage on a *dense* lattice, consistent with that literature’s own scope of sparse dopants and adatoms. The limit is structural, not a tuning artefact. Second, **on noiseless data a good peak-picker is competitive on the heavy atoms**. We report this rather than a flattering weaker baseline: with no shot noise a raw local maximum is already a clean estimator of an isolated bright column, so the margin quoted here is a lower bound on the method’s advantage, and the discriminating comparison is at finite dose (Sec. III D). What a peak-picker cannot provide at any dose is a species label or a calibrated error bar.

C. Calibrated uncertainty

Figure 3(a) quantifies why a per-peak bound is inadequate here. The variance inflation factor of the joint fit is 1.04 at the 3.9 Å heavy-atom spacing — where atoms are effectively independent — but 2.02 in depth (and 2.28 in amplitude) at the 1.95 Å Ti–O spacing, so a per-peak Cramér–Rao bound understates σ_z by $\approx 1.4\times$ for every apical oxygen. Since 85% of atoms in the field have a neighbour within 2.5 Å, the per-peak form is optimistic almost everywhere. Inverting the full Fisher matrix restores the inflation per atom automatically, with no tuned constant.

Even so, the raw model σ treated as Gaussian is badly miscalibrated on near-noiseless simulation, covering only $\approx 8\%$ of atoms at a nominal 68% (Fig. 3b): with no shot noise the residual-based variance is near zero while the true error is dominated by systematics the residual cannot see. Conformal calibration supplies the missing scale empirically. Held out on a 50/50 split, the conformal interval tracks the ideal diagonal across the whole range of nominal coverage, reaching 69/69/69% in (x, y, z) at a 68% target and 96/96/96% at 95%. The Mondrian stratification is not cosmetic: the per-stratum quantiles $q_{68}(z)$ range from 6.7 to 14.4, a factor of 2.2, so a single pooled constant necessarily leaves some strata conservative and others optimistic — visible in Fig. 3(c), where pooling drops the worst stratum to 84% against a 95% target while the label-conditional quantile restores every stratum to 95–100%.

The decisive test of an uncertainty estimate is whether it survives a change of dataset. We applied the identical code, with no tuned constants, to an independent reconstruction of the same material at a different in-plane sampling (0.10 against 0.15 Å), a different slice discretisation (105 against 70 layers), roughly half the phase amplitude and real reconstruction noise. Re-running the conformal step on that volume’s own atoms gives 69/69/69% at the 68% target and 96/96/96% at 95% — coverage as good as on the volume the method was developed on. For comparison, the hand-tuned per-species error floors used in an earlier version of this work, calibrated to 68% coverage on the first volume, transferred to 38/47/56% on the second. The per-stratum quantiles on the harder volume span an even wider range (10.9 to 23.3), which is exactly the regime a single floor cannot serve and conformal handles for free.

Three limitations should be stated plainly. The interval is a *localisation* interval, conditional on the atom having been correctly detected and typed; a spurious detection or a species swap is a different failure mode, reported separately as precision (0.97) and confusion (1.1%). The species posterior we export is a within-column amplitude confidence and is not itself calibrated, though it concentrates the residual errors usefully: labels below 0.9 are enriched for mislabelling by more than an order of magnitude over the base rate. And conformal calibration assumes exchangeability between calibration and test atoms, which is why the workflow re-runs the calibration per volume and exports the per-stratum coverage so that a failure is visible.

D. Behaviour at finite dose

Table III runs the same configuration over a dose series on the independent reconstruction. Performance is maintained to $10^8 \text{ e} \text{ \AA}^{-2}$ and collapses below 10^6 . The residual gap to the coherent reference (Ti 74% against 89%) is not attributable to the localisation constants but to a genuinely harder reconstruction: a 2.0 Å axial ker-

dose ($\text{e} \text{ \AA}^{-2}$)	precision	Pb	Ti	O	confusion
10^{10}	0.83	86 %	74 %	63 %	13.7 %
10^8	0.87	87 %	72 %	67 %	11.8 %
10^6	0.95	75 %	25 %	7 %	38.0 %
10^4	—	0 %	0 %	0 %	—

TABLE III. Dose series on an independent reconstruction (105 layers, 0.10 Å sampling), run with the *same* configuration as the coherent reference. The 10^6 row is the case in point for precision being uninformative. A guided-detection deduplication guard added subsequently reduces the 10^{10} confusion to 12.4% with titanium recall unchanged.

nel against 1.0 Å, and depth-registration correlation 0.44 against 0.63.

Two lessons generalise. First, **precision is not a safety indicator**. At $10^6 \text{ e} \text{ \AA}^{-2}$ precision reads 0.95 — apparently healthy — while titanium recall is 25%, oxygen 7% and 38% of species labels are wrong: the surviving bright lead atoms are correctly placed, and precision measures only them. Species confusion and σ -coverage are the diagnostics that fire, and the released pipeline prints both as health warnings. Second, **species assignment is coupled to depth registration**. A 1 Å depth error on a lattice whose Ti and O alternate every 1.95 Å swaps labels wholesale; correcting the depth offset from 1.26 to 0.29 Å took titanium recall from 53% to 74% and confusion from 19.0% to 13.7% at $10^{10} \text{ e} \text{ \AA}^{-2}$. Sub-ångström depth accuracy is not merely a precision figure; it is a prerequisite for correct chemistry.

E. Polarisation with error bars

We now close the loop to the physical observable. Evaluating Eq. (1) on the located atoms, 72% of bulk titanium acquire a complete oxygen cage from blindly located oxygens alone; 171 of these match a ground-truth titanium and are scored in Fig. 4.

The in-plane off-centring is recovered essentially exactly. The median error in δ_x is 0.005 Å and in δ_y 0.004 Å, against a true in-plane off-centring of mean magnitude 0.274 Å; the recovered mean magnitude is 0.273 Å. As a vector the in-plane off-centring has median error 0.007 Å and 90th percentile 0.017 Å, and its *direction* — the quantity a vortex map is made of — has median error 0.9°, with 95% of titanium within 30°. The correlation between recovered and true δ_x is 0.96 over all matched atoms and 1.00 once a small tail is excluded.

That tail is the honest caveat. Five per cent of titanium have a cage of the wrong composition — a missing or misassigned oxygen displaces the centroid — and for these the in-plane error jumps to a median of 0.45 Å, comparable to the signal itself. Crucially, the propagated positional interval does *not* cover them: it covers 99% of the well-caged titanium but only 12% of the tail. This is not a failure of the calibration but a statement of its

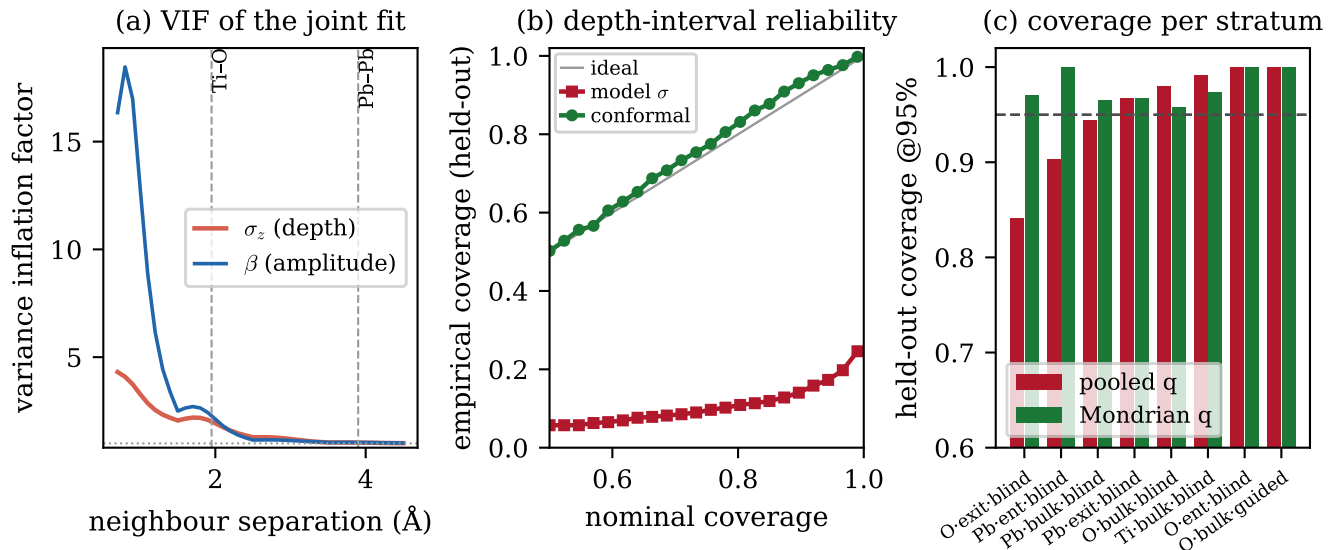


FIG. 3. Calibrated uncertainty. (a) Variance inflation factor of the joint fit against neighbour separation: near unity at the 3.9 Å heavy-atom spacing but ≈ 2 in depth (and larger in amplitude) at the 1.95 Å Ti-O spacing, so a per-peak Cramér–Rao bound is optimistic wherever responses overlap. (b) Held-out reliability of the depth interval: the raw model standard error treated as Gaussian (red) severely under-covers on near-noiseless data, while the conformal interval (green) tracks the ideal diagonal. (c) Held-out coverage at the 95% level for the strata where a single pooled quantile (red) departs most from target; the label-conditional quantile (green) restores coverage in each. Split-conformal, 50/50 calibration/test split.

scope, and it is the same caveat as in Sec. II E propagated forward. The interval is conditional on correct detection; a gross error in cage *composition* is a detection failure, and it must be reported through the detection metrics (precision, confusion, cage completeness) rather than absorbed into a positional error bar. A practical map should therefore carry both, and the 28% of titanium with incomplete cages should be shown as absent rather than interpolated.

The along-beam component tells the opposite story, and it is the more consequential result. Its true spread across the scanned volume is small ($\sigma = 0.074$ Å, since in this zone axis the vortex polarisation is predominantly in-plane), while the propagated uncertainty on δ_z is 0.237 Å — more than three times the entire signal. The measured correlation with truth is correspondingly poor ($r = 0.42$, median error 0.122 Å). **The along-beam component of the polarisation is not measured by this experiment.** The value of the calibrated uncertainty is that it says so *blind*: comparing the propagated σ_z with the spread of the recovered δ_z requires no ground truth, and it correctly declares the along-beam map uninformative while certifying the in-plane map to 0.01 Å. Propagated intervals cover 95/93/99% of true errors in (x, y, z) at a 95% target, so the uncertainties are honest in both directions — neither the confidence in-plane nor the pessimism along the beam is overstated.

This is a quantitative statement of the limitation that motivated the second half of the project. Depth *localisation* of individual atoms is good — 0.37 Å RMS, comfort-

ably better than the ≈ 2 Å depth resolution, in the same way that single-molecule localisation beats the diffraction limit [46]. But a polarisation is a *difference* of two such coordinates, and differencing two 0.3 Å depth errors cannot resolve a 0.07 Å displacement. In-plane the same argument runs the other way: the in-plane error is 0.03 Å against a 0.27 Å signal, so the map is quantitative.

F. Can EELS see the missing component?

The second phase asks whether the along-beam polarisation, invisible to a projected or depth-differenced position measurement, leaves a spectroscopic fingerprint. The calculation was built as a validated staircase, each step gated before the next.

a. Validation. CASTEP/PBE reproduces the ferroelectric instability: single-point energies across a displacement series $\mathbf{u} = s \mathbf{u}_0$ at fixed tetragonal strain place the fully polar structure 203 meV per formula unit below the centrosymmetric reference, monotonically, with the descent flattening towards $s = 1$ so that the experimental displacement sits at the PBE minimum. The core-hole recipe was then fixed by reproducing a known edge: the rutile TiO₂ O K ELNES comes out with its textbook shape, the t_{2g}/e_g near-edge peaks split by 2.7 eV (experiment 2.5–3 eV) with $e_g > t_{2g}$, followed by the broad O 2p–Ti 4sp band. As an artefact check, the dichroism of the *cubic* cell computed at the isotropic Ti site is 0.02% — the two spectra are identical, as symmetry demands,

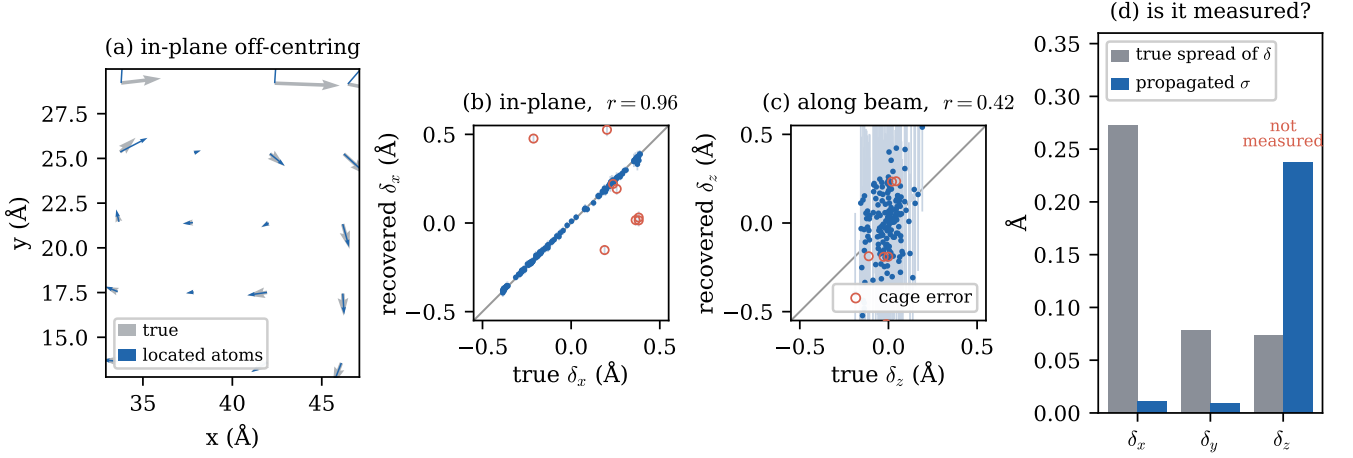


FIG. 4. Local polarisation from the located atoms. (a) In-plane Ti-O₆ off-centring averaged per atomic column: recovered (blue) over ground truth (grey). (b) Recovered against true in-plane off-centring, with propagated 95% intervals; open red circles mark the 5% of titanium whose oxygen cage has the wrong composition. (c) The same for the along-beam component. (d) The propagated uncertainty against the true spread of each component: in-plane the uncertainty is a small fraction of the signal, along the beam it exceeds it — a ground-truth-free statement that the along-beam polarisation is not determined by this measurement.

so any dichroism elsewhere is physical rather than numerical.

b. The signal. For the tetragonal cell with its polar axis along the beam, the O K edge is strongly dichroic: $\max |\Delta| / \max S = 78\%$ and $\int |\Delta| / \int S = 47\%$ (Fig. 5). The origin is a clean σ/π anisotropy: with $\mathbf{q} \perp c$ the edge shows a sharp π^* peak at ≈ 8.5 eV above onset from O $2p_{x,y}$, while with $\mathbf{q} \parallel c$ it shows σ^* features at ≈ 11 and 18.5 eV from the O $2p_z$ pointing along the Ti-O-Ti chain. Orientation of the polar axis relative to the beam therefore imprints strongly on the O K edge, far above any detectability floor.

c. Decomposition. A large orientation dichroism is not by itself a polarisation measurement, because the tetragonal backbone is anisotropic even without off-centring. Holding the lattice parameters fixed and varying only the internal displacement s separates the two (Fig. 6). At $s = 0$ — zero polarisation, full tetragonal strain — the dichroism is already 70.1%; at $s = 1$ it is 78.2%. So roughly 70 of the 78 points are the strain-induced backbone σ/π anisotropy, and the ferroelectric off-centring's own signature is the remaining, monotonic contribution: an 18.7% change in the along-beam-probed O K spectrum between $s = 0$ and $s = 1$ (4.3% at $s = 0.5$), plus 8 points of additional dichroism. This is the first-principles counterpart of the empirical observation that the O K edge responds to titanium off-centring [44], and it is what makes the measurement a polarisation probe rather than merely a strain probe.

d. Detectability. The intrinsic dichroism must then survive the acquisition geometry. For the O K edge at 300 keV, $\theta_E = 1.09$ mrad and the magic angle is 4.32 mrad $= 3.98 \theta_E$, reproducing the textbook factor of ≈ 4 and validating our aperture model. Folding the in-

trinsic anisotropy through the collection geometry gives a surviving fraction of 0.62 at $\beta = 1$ mrad, 0.28 at 2 mrad, -0.04 at 5 mrad and -0.33 at 100 mrad. The 100 mrad convergence of the ptychography probe therefore lies far past the magic angle and averages the anisotropy away — indeed inverts its sign. A combined ptychography-plus-EELS experiment using one pixelated detector with a central hole would collect at $\beta \approx 100$ mrad and wash out precisely the signal it was built to measure. Recovering the along-beam polarisation requires a dedicated small EELS aperture, $\lesssim 2$ mrad; at a 2% fractional anisotropy this needs $\approx 4.5 \times 10^4$ counts per channel for SNR 3, which sets the dose target.

e. Two intrinsic limits. First, the dipole ELNES is even in $\mathbf{q}$, so it measures the *magnitude* of the along-beam polarisation but not its sign; distinguishing $+P$ from $-P$ needs beyond-dipole or off-axis momentum transfer. A dynamical multislice core-loss simulation of up- and down-polarised domains confirms this directly: the two give O K spectra identical to within 1.0%. Second, in the actual labyrinth viewed along this zone axis the polarisation is mostly in-plane — over 2560 polar titanium, the median angle from the beam is 82° (interquartile range 73 – 87°), an along-beam fraction of only ≈ 0.14 . The full along-beam case computed here is therefore an upper bound for this orientation; a different zone axis would raise the along-beam share, and the technique is best suited to geometries chosen for it.

IV. CONCLUSIONS AND FURTHER WORK

Fitting a reconstructed multislice-ptychographic volume as a superposition of a *measured* single-

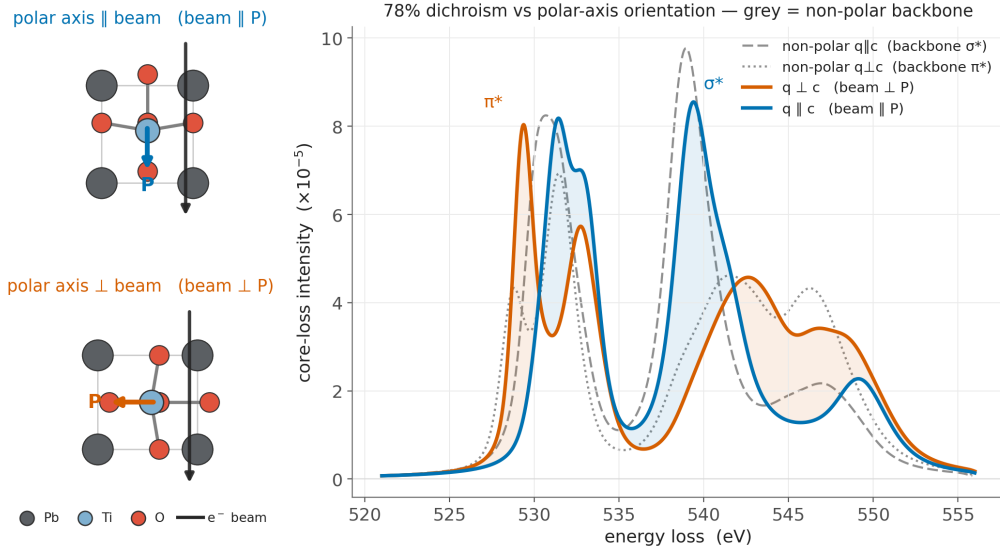


FIG. 5. Orientation dichroism of the PbTiO_3 O K edge from core-hole CASTEP + OptaDOS. With the momentum transfer across the polar axis ($\mathbf{q} \perp c$, beam $\perp \mathbf{P}$) the edge shows a sharp π^* peak; along it ($\mathbf{q} \parallel c$, beam $\parallel \mathbf{P}$) it shows the σ^* features of the O $2p_z$ states directed along the Ti–O–Ti chain. Grey curves are the same two orientations for the zero-polarisation (non-polar, equally strained) reference, isolating the backbone contribution.

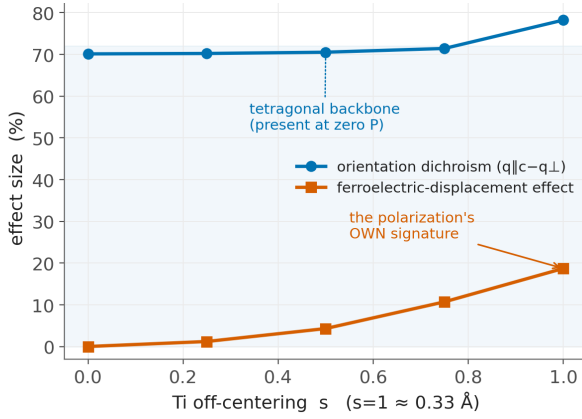


FIG. 6. Separating strain from polarisation. Holding the lattice parameters fixed and varying only the internal Ti off-centring s , the total orientation dichroism (blue) is already 70% at zero polarisation — the tetragonal backbone — while the ferroelectric displacement’s own signature (orange) grows monotonically to 19% at the experimental off-centring $|\mathbf{u}| \approx 0.33 \text{ \AA}$.

atom response recovers the oxygen sublattice of a $\text{PbTiO}_3/\text{SrTiO}_3$ superlattice where the established deconvolve-then-detect workflow does not: 96% bulk recall against at most 59%, with the entire margin concentrated in the axially overlapped oxygen (82% against 9–26%) that a single-orientation measurement buries under its titanium neighbour. The advantage is not a tuning artefact — deconvolution’s limit is the axial spatial frequencies the aperture never records — and it comes with two things no peak detector provides: species labels

(1.1% confusion) and calibrated per-atom uncertainty.

The uncertainty machinery is, we think, the more transferable contribution. Combining a *joint* Cramér–Rao bound — which recovers the twofold variance inflation that a per-peak bound misses wherever responses overlap, and 85% of atoms here have an overlapping neighbour — with Mondrian conformal calibration produces intervals that hold their nominal coverage per stratum, contain no hand-tuned constant, and transfer unchanged to an independent reconstruction at different sampling, slicing and dose, where hand-tuned error floors fail badly (at a 68% target, 69% coverage against 38–56%). We would encourage the reporting of coverage rather than of a fitted σ : the same pipeline that reads σ -calibrated at one dose reads 38% species confusion at another while its precision still reads 0.95.

Propagated into the physics, the result is sharply two-sided. The in-plane polarisation map is quantitative — median error $0.007 \text{ \AA}$ in magnitude and 0.9° in direction, against a $0.27 \text{ \AA}$ signal — while the along-beam component is not measured at all, because differencing two $0.3 \text{ \AA}$ depth coordinates cannot resolve a $0.07 \text{ \AA}$ displacement. The calibrated uncertainty establishes this without ground truth, which is precisely what such machinery is for. First-principles core-hole DFT then shows the missing component is spectroscopically accessible in principle: the O K edge is 78% dichroic to the polar-axis orientation, of which a monotonic 19% is the displacement’s own signature rather than the tetragonal backbone’s, though the dipole limit forbids recovering the *sign* and the magic angle forbids using the 100 mrad ptychography aperture to do it.

Several directions follow. *Method*: the one-kernel-

for-all-species assumption is weakest where it matters most, since oxygen’s larger Debye–Waller factor genuinely broadens its response under thermal scattering; in-situ vacancy-difference kernels — the lattice reconstructed with and without a single atom — would supply the in-crystal oxygen matched filter no existing kernel provides. The 5% cage-composition tail of Sec. III E, the dominant residual error on the polarisation map, is a detection rather than a localisation problem and needs a calibrated cage-completeness criterion; learned denoising of the volume before fitting [47] is a natural route below $10^6 \text{ e} \text{ \AA}^{-2}$. *Physics*: the outstanding EELS cross-checks are the rotational-invariance test and the multiplicity-weighted full O K edge; beyond them, a faithful vortex simulation would inject a *per-atom* $\sigma(E)$ keyed to each oxygen’s local polar-axis orientation, closing the loop from the spectroscopy back onto the imaged sample. *Instrument*: the clearest experimental conclusion is a design constraint — a combined depth-resolved-imaging and polarisation-spectroscopy experiment cannot share one large aperture. Whether the two can be fused into a single three-dimensional vector polarisation measurement, along the lines of energy-filtered ptychography of the oxygen sublattice [11] or momentum-selective EELS of these very flux-closure structures [48], is the question this work leaves open.

V. REPRODUCIBILITY

All code, the input structure and the per-run parameters are in the project repository, and the pipeline is scripted end to end: simulation (`sim/simulate_4dstem.py`), reconstruction (`PtychoShelves GPU MS`), localisation (`atomfind/run_atomfind.py`), polarisation (`atomfind/polarisation.py`) and the DFT staircase (`eels/build_cells.py`, `run_coreloss.slurm`, `analyze_elnes.py`), with a unit-test suite guarding the structure generation and parsing.

a. The peer-reproducible result. The nominated result is the localisation-with-UQ output of Sec. III E. A peer receives the reconstructed phase volume (a $70 \times 404 \times$

404 array), the measured kernel, and the script; running `run_atomfind.py` followed by `polarisation.py` produces `found.atoms.csv` — element, (x, y, z) in ångström, per-axis 95% conformal half-widths, model σ , species confidence, and a flag marking lattice-constrained detections — together with the conformal calibration table and the Ti–O₆ off-centring with propagated uncertainty. The run takes a few minutes on one core and needs no GPU, so it is well within the moderate resources specified; the upstream simulation and reconstruction, which are not, are supplied pre-computed.

b. Expected values and their uncertainty. A peer should reproduce, bit-for-bit on the same input, 1834 atoms with precision 0.97; bulk recall 96/96/96% for Pb/Ti/O; species confusion 1.1%; in-plane RMS accuracy 0.032 Å and depth RMS 0.37 Å. The uncertainty on each reported coordinate is the exported conformal half-width, whose validity is the coverage statement itself: 96% of true errors fall inside the 95% interval, per stratum, verified held-out on a 50/50 split. Typical values are 0.02 Å in-plane and 0.5 Å in depth, and they are strongly heteroscedastic — up to 1.5 Å in depth for weak oxygen near the exit surface — which is why the per-atom interval, not a global figure, is the quantity to propagate. For the derived polarisation, the in-plane off-centring carries a propagated σ of 0.011 Å and the along-beam component 0.237 Å. The single stochastic element is the conformal calibration split; it is seeded, and re-seeding changes the quantiles by less than 5%. The known failure modes a peer should expect to see flagged are the 28% of titanium with incomplete oxygen cages and the 5% with cage-composition errors, both reported by the script rather than silently absorbed.

ACKNOWLEDGMENTS

The relaxed PbTiO₃/SrTiO₃ supercell was provided by collaborators and is used with their agreement; we thank them for the structure. Calculations were performed on the Warwick Blythe cluster and the Scientific Computing RTP. TM is supported by the EPSRC HetSys Centre for Doctoral Training (EP/S022848/1).

-
- [1] A. K. Yadav, C. T. Nelson, S. L. Hsu, Z. Hong, J. D. Clarkson, C. M. Schlepütz, A. R. Damodaran, P. Shafer, E. Arenholz, L. R. Dedon, D. Chen, A. Vishwanath, A. M. Minor, L. Q. Chen, J. F. Scott, L. W. Martin, and R. Ramesh, *Nature* **530**, 198 (2016).
 - [2] Y. L. Tang, Y. L. Zhu, X. L. Ma, A. Y. Borisevich, A. N. Morozovska, E. A. Eliseev, W. Y. Wang, Y. J. Wang, Y. B. Xu, Z. D. Zhang, and S. J. Pennycook, *Science* **348**, 547 (2015).
 - [3] S. Das, Y. L. Tang, Z. Hong, M. A. P. Gonçalves, M. R. McCarter, C. Klewe, K. X. Nguyen, F. Gómez-Ortiz, P. Shafer, E. Arenholz, V. A. Stoica, S.-L. Hsu, B. Wang, C. Ophus, J. F. Liu, C. T. Nelson, S. Saremi, B. Prasad, A. B. Mei, D. G. Schlom, J. Íñiguez, P. García-Fernández, D. A. Muller, L. Q. Chen, J. Junquera, L. W. Martin, and R. Ramesh, *Nature* **568**, 368 (2019).
 - [4] J. Junquera, Y. Nahas, S. Prokhorenko, L. Bellaiche, J. Íñiguez, D. G. Schlom, L.-Q. Chen, S. Salahuddin, D. A. Muller, L. W. Martin, and R. Ramesh, *Rev. Mod. Phys.* **95**, 025001 (2023).
 - [5] S. Gao, P. Wang, F. Zhang, G. T. Martinez, P. D. Nellist, X. Pan, and A. I. Kirkland, *Nature Communications* **8**, 163 (2017).

- [6] Z. Chen, Y. Jiang, Y.-T. Shao, M. E. Holtz, M. Odstrčil, M. Guizar-Sicairos, I. Hanke, S. Ganschow, D. G. Schlom, and D. A. Muller, *Science* **372**, 826 (2021).
- [7] A. M. Maiden, M. J. Humphry, and J. M. Rodenburg, *Journal of the Optical Society of America A* **29**, 1606 (2012).
- [8] P. D. Nellist, B. C. McCallum, and J. M. Rodenburg, *Nature* **374**, 630 (1995).
- [9] Y. Jiang, Z. Chen, Y. Han, P. Deb, H. Gao, S. Xie, P. Purohit, M. W. Tate, J. Park, S. M. Gruner, V. Elser, and D. A. Muller, *Nature* **559**, 343 (2018).
- [10] K. X. Nguyen, Y. Jiang, C.-H. Lee, P. Kharel, Y. Zhang, A. M. van der Zande, and P. Y. Huang, *Science* **383**, 865 (2024).
- [11] Z. Dong, M. Huo, J. Li, J. Li, P. Li, H. Sun, L. Gu, Y. Lu, M. Wang, Y. Wang, and Z. Chen, *Nature* **630**, 847 (2024).
- [12] Z. Dong, Y. Zhang, C.-C. Chiu, S. Lu, J. Zhang, Y.-C. Liu, S. Liu, J.-C. Yang, P. Yu, Y. Wang, and Z. Chen, *Nat. Commun.* **16**, 1219 (2025).
- [13] Y. Yang, C.-C. Chen, M. C. Scott, C. Ophus, R. Xu, A. Pryor, L. Wu, F. Sun, W. Theis, J. Zhou, M. Eisenbach, P. R. C. Kent, R. F. Sabirianov, H. Zeng, P. Ercius, and J. Miao, *Nature* **542**, 75 (2017).
- [14] Y. Yang, J. Zhou, F. Zhu, Y. Yuan, D. J. Chang, D. S. Kim, M. Pham, A. Rana, X. Tian, Y. Yao, S. J. Osher, A. K. Schmid, L. Hu, P. Ercius, and J. Miao, *Nature* **592**, 60 (2021).
- [15] R. Xu, C.-C. Chen, L. Wu, M. C. Scott, W. Theis, C. Ophus, M. Bartels, Y. Yang, H. Ramezani-Dakhel, M. R. Sawaya, H. Heinz, L. D. Marks, P. Ercius, and J. Miao, *Nature Materials* **14**, 1099 (2015).
- [16] A. Ishizuka, K. Ishizuka, R. Ishikawa, N. Shibata, Y. Ikuhara, H. Hashiguchi, and R. Sagawa, *Microscopy* **70**, 241 (2021).
- [17] A. De Backer, K. H. W. van den Bos, W. Van den Broek, J. Sijbers, and S. Van Aert, *Ultramicroscopy* **171**, 104 (2016).
- [18] J. M. Cowley and A. F. Moodie, *Acta Crystallographica* **10**, 609 (1957).
- [19] E. J. Kirkland, *Advanced Computing in Electron Microscopy*, 2nd ed. (Springer, New York, 2010).
- [20] J. Madsen and T. Susi, *Open Research Europe* **1**, 24 (2021).
- [21] I. Lobato and D. Van Dyck, *Acta Crystallographica Section A* **70**, 636 (2014).
- [22] R. F. Loane, P. Xu, and J. Silcox, *Acta Crystallographica Section A* **47**, 267 (1991).
- [23] L.-M. Peng, G. Ren, S. L. Dudarev, and M. J. Whelan, *Acta Crystallographica Section A* **52**, 456 (1996).
- [24] J. M. Rodenburg and H. M. L. Faulkner, *Applied Physics Letters* **85**, 4795 (2004).
- [25] K. Wakonig, H.-C. Stadler, M. Odstrčil, E. H. R. Tsai, A. Diaz, M. Holler, I. Usov, J. Raabe, A. Menzel, and M. Guizar-Sicairos, *Journal of Applied Crystallography* **53**, 574 (2020).
- [26] P. Thibault and M. Guizar-Sicairos, *New Journal of Physics* **14**, 063004 (2012).
- [27] M. Odstrčil, A. Menzel, and M. Guizar-Sicairos, *Optics Express* **26**, 3108 (2018).
- [28] P. Thibault and A. Menzel, *Nature* **494**, 68 (2013).
- [29] M. Odstrčil, P. Baksh, S. A. Boden, R. Card, J. E. Chad, J. G. Frey, and W. S. Brocklesby, *Optics Express* **24**, 8360 (2016).
- [30] S. J. Clark, M. D. Segall, C. J. Pickard, P. J. Hasnip, M. I. J. Probert, K. Refson, and M. C. Payne, *Zeitschrift für Kristallographie - Crystalline Materials* **220**, 567 (2005).
- [31] A. J. Morris, R. J. Nicholls, C. J. Pickard, and J. R. Yates, *Computer Physics Communications* **185**, 1477 (2014).
- [32] S. G. Mallat and Z. Zhang, *IEEE Transactions on Signal Processing* **41**, 3397 (1993).
- [33] J. A. Högbom, *Astron. Astrophys. Suppl. Ser.* **15**, 417 (1974).
- [34] R. Ishikawa, A. R. Lupini, S. D. Findlay, T. Taniguchi, and S. J. Pennycook, *Nano Letters* **14**, 1903 (2014).
- [35] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World* (Springer, New York, 2005).
- [36] A. N. Angelopoulos and S. Bates, *Foundations and Trends in Machine Learning* **16**, 494 (2023).
- [37] V. Vovk, D. Lindsay, I. Nouretdinov, and A. Gammerman, *Mondrian confidence machine*, Technical Report (Royal Holloway, University of London, 2003).
- [38] R. F. Egerton, *Electron Energy-Loss Spectroscopy in the Electron Microscope*, 3rd ed. (Springer, New York, 2011).
- [39] M. Nelhiebel, P.-H. Louf, P. Schattschneider, P. Blaha, K. Schwarz, and B. Joffrey, *Physical Review B* **59**, 12807 (1999).
- [40] B. Joffrey, P. Schattschneider, and C. Hébert, *Ultramicroscopy* **102**, 61 (2004).
- [41] S.-P. Gao, C. J. Pickard, M. C. Payne, J. Zhu, and J. Yuan, *Physical Review B* **77**, 115122 (2008).
- [42] R. J. Nicholls, A. J. Morris, C. J. Pickard, and J. R. Yates, *Journal of Physics: Conference Series* **371**, 012062 (2012).
- [43] A. Torres-Pardo, A. Gloter, P. Zubko, N. Jecklin, C. Lichtensteiger, C. Colliex, J.-M. Triscone, and O. Stéphan, *Physical Review B* **84**, 220102(R) (2011).
- [44] M. Bugnet, G. Radtke, S. Y. Woo, G.-Z. Zhu, and G. A. Botton, *Phys. Rev. B* **93**, 020102(R) (2016).
- [45] T. Mizoguchi, I. Tanaka, S.-P. Gao, and C. J. Pickard, *J. Phys.: Condens. Matter* **21**, 104204 (2009).
- [46] R. E. Thompson, D. R. Larson, and W. W. Webb, *Biophys. J.* **82**, 2775 (2002).
- [47] Y. Wu, Z. Huang, H. Chen, B. He, M. Qi, H. Zheng, P. Zhao, S. Jia, and J. Wang, *Ultramicroscopy* **10.1016/j.ultramic.2026.114166** (2026).
- [48] Y. Huang, F. Gong, X. Yan, T. Aoki, and X. Pan, *Microsc. Microanal.* **32**, ozag053.780 (2026).